

Technical Datasheet — V10a_final_h45 (extreme-cold equipment battery)

Field	Value	Source
Rated capacity	5.0 Ah nominal (parameter set); 5.0618 Ah simulation-verified at 1C	Chen2020 set; r6_final_1c_dfn.json:capacity_ah
Nominal voltage / window	3.89 V discharge midpoint; 2.5 – 4.2 V	r6_final_energy_dfn.json:midpoint_voltage_v; Chen2020 cut-offs
Rated energy	18.400 Wh (V·I time integration of 1C discharge)	r6_final_energy_dfn.json:energy_wh
Energy density	564.84 Wh/kg (contract caliber, electrolyte excluded from mass)	r6_final_energy_dfn.json:energy_density_w_h_kg
Volumetric energy density	969.50 Wh/L (layer-stack volume, electrolyte/casing excluded)	r6_final_energy_dfn.json:energy_density_w_h_l
DC resistance	3.367 mOhm (OCV - V@10% discharge over I_1C)	r6_final_energy_dfn.json:dcr_ohm
Maximum continuous discharge rate	1C (simulation protocol; higher rates not simulated)	r6_final_1c_dfn.json
Fast-charge capability	4C at 45 C: T_max 329.33 K (limit 333.15 K), no lithium plating (anode min +28.96 mV)	r6_final_4C45_safety_dfn.json
Low-temperature performance	-20 C 1C retention 0.9944 (5.0337 Ah)	derived_r6_final_dfn.json
Operating temperature range	simulated at -20 C and 45 C ambient; intermediate temperatures not simulated	lowT_discharge / 4C_charge_45C protocols
Cycle life	Not simulated (requires aging model) — not fabricated	
Safety determination	no plating at 4C/45C; max temperature below 333.15 K red line	r6_final_4C45_safety_dfn.json
Dimensions and mass	electrode sheet 65 mm x 1580 mm; stack thickness 184.8 um; mass 32.58 g (contract caliber, electrolyte excluded; +7.98 g electrolyte estimate)	Chen2020 set; r6_final_energy_dfn.json
Cooling requirement	total heat transfer coefficient 45 W/m2K (design value)	params_r6_v10a_h45.json